

Mohit Singhal

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PROFESSIONAL SUMMARY

Software Engineer specializing in **Generative AI, Machine Learning, LLMs**, and enterprise software development. Experienced in designing and developing scalable AI-powered software solutions using Python, LangChain, LangGraph, REST APIs, and cloud platform services. Passionate about building **innovative** AI products that drive business **growth**. Strong analytical, problem-solving, and algorithmic thinking with experience **collaborating** in fast-paced, cross-functional Agile teams while following software engineering best practices. Microsoft Certified: Azure AI Fundamentals (AI-900).

EXPERIENCE

CodeStore Technologies Pvt. Ltd., Noida (On-Site)

June 2025 – Present

Software Engineer (Trainee → Full-time)

- Led the technical implementation of Microsoft Copilot Studio agents for HR and departmental use cases while collaborating with stakeholders to automate routine HR queries by approximately 30%.
- Developed AI-powered full stack applications, backend services, REST APIs, automation pipelines, and UI components using Python, React, and cloud platform services. Built scalable microservices, workflow automation solutions, dashboards, and web scraping applications while following containerization-ready deployment practices.
- Conducted AI research, software validation, QA testing, troubleshooting, documentation, and PoC development while collaborating with cross-functional engineering teams in fast-paced Agile software development environments, integrating Microsoft 365, Teams, Outlook, Zoho, and Azure DevOps platform services.
- Collaborated with business stakeholders and technical teams to gather requirements, design scalable AI solutions, and align implementations with business objectives and operational requirements.
- Designed and maintained clean, scalable automation infrastructure and software pipelines integrating APIs, internal tools, and business systems while following software engineering best practices, improving application reliability, deployment efficiency, monitoring, performance optimization, and operational performance.

SKILLS

- **Automation & AI:** n8n, Microsoft Copilot Studio, Power Automate, Power Apps, Prompt Engineering, RAG, AI Agents, LangChain, LangGraph, OpenAI, Gemini
- **Backend Development:** FastAPI, REST APIs, Webhooks, MCP Server
- **Cloud & Platform Services:** Microsoft Azure, AWS Fundamentals, Azure DevOps
- **Databases:** PostgreSQL, SQL, Pinecone, Vector Databases, Supabase
- **Version Control:** Git, GitHub, Azure DevOps
- **Professional Skills:** Stakeholder Collaboration, Teamwork, Adaptability, Technical Documentation

PROJECTS

AI-Whatsapp Customer Support Agent

- **Tools/Tech Stack:** n8n, PostgreSQL, Pinecone, WATI API, Webhooks, GPT-4.1-mini, Google Sheets, Gmail, Telegram
- **Agent Overview:** Designed and developed a scalable AI-powered software platform comprising Customer Support, Follow-up, and Lead Qualification agents using conversational AI, Machine Learning concepts, Prompt Engineering, and LLMs to automate customer interactions and reduce manual lead qualification effort by approximately 25%.
- **RAG Pipeline:** Built an end-to-end RAG pipeline by performing documentation chunking, generating embeddings, and enabling semantic retrieval for accurate product-related query resolution via text-embedding-3-large model.
- **Agent Design:** Implemented dynamic follow-up workflows and prompt-engineered lead qualification to classify leads based on conversation history, while leveraging shared context memory for coordinated agent execution.
- **Monitoring, Validation, Security & Reliability:** Implemented AI-to-human handoff workflows, conversation monitoring, validation mechanisms, security controls, reliability controls, error alerting, and collaboration workflows for production AI systems, improving operational efficiency and reducing support costs by 40%.

DeepResearch Agent — Autonomous Multi-Agent Research Pipeline [↗](#)

- **Tech Stack:** Python, LangChain, LangGraph, Gemini, Tavily API, BeautifulSoup, Streamlit, LLM orchestration
- **Agent Overview:** Built an autonomous AI-powered research platform using LangChain and LangGraph, implementing scalable software architecture, Agentic AI, and Machine Learning concepts for end-to-end research, report generation, critique, and autonomous decision making.
- **Workflow Design:** Designed a 4-stage Agentic Architecture comprising Search Agent, Reader Agent, Writer Chain, and Critic Chain using LCEL and Gemini 2.5 Flash, enabling autonomous task orchestration and tool-based reasoning.
- **Research Automation:** Implemented tool-using agents with web search and scraping capabilities for autonomous information retrieval.
- **Deployment:** Built and deployed a Streamlit dashboard for real-time agent monitoring.

AI-Powered Employee Assessment Agent [↗](#)

- **Tech Stack:** Microsoft Copilot Studio, Power Automate, GPT-4.1-mini, Zoho People, ZET CLI SDK, Azure DevOps Wiki APIs, Microsoft Graph APIs, React, Teams, Outlook, REST APIs, Power Apps
- **Agent Overview:** Developed a RAG-based employee assessment platform using Microsoft Copilot Studio that generates dynamic policy assessments from HR, Sales, and departmental wikis through Azure DevOps Wiki ,JSON data and Microsoft Graph API integrations.
- **Documentation Automation:** Built documentation-driven content creation, validation, and compliance workflows using Azure DevOps Wiki APIs, Microsoft Graph APIs, JSON data sources, and RAG architecture to automate policy assessment generation and knowledge management processes.
- **Workflow & Integrations:** Built CRM and HR system integrations using Zoho People, Microsoft Graph APIs, REST APIs, and Power Automate for assessment workflows and automated notifications.
- **Assessment Engine:** Implemented AI-driven content generation and validation workflows, dynamic assessment sizing based on wiki complexity, semantic retrieval, contextual answer explanations, and policy-grounded responses using GPT-4.1-mini.
- **Analytics & Dashboard:** Developed React-based admin dashboards focused on improving UX (User Experience), employee performance management, and AI-generated performance summaries while storing assessment data in Zoho People for compliance reporting and knowledge tracking.

CERTIFICATIONS

- **Microsoft Certified: Azure AI Fundamentals (AI-900)** [↗](#)
- **AWS Cloud Practitioner - Udemy** [↗](#)

ADDITIONAL TECHNICAL EXPERIENCE

- **Algorithmic Trading Infrastructure:** Developing an end-to-end algorithmic trading platform using Python, FastAPI, Flask, Upstox Broker APIs, OAuth 2.0 Authentication, REST APIs, WebSockets, Protocol Buffers (Protobuf), AsyncIO, Azure Virtual Machines, Linux (Ubuntu), Nginx, Git/GitHub, MobaXterm, and Cron Jobs. Built backend services for secure broker authentication, live market data ingestion, automated order execution, position management, trailing stop-loss, scheduled auto square-off, risk management, production monitoring, and cloud deployment.
- **Quantitative Strategy Research & Backtesting:** Researching, designing, testing, and optimizing systematic trading strategies across equity and derivatives markets, including Range Breakout, ATR-based momentum, Market Depth analysis, option buying, option selling, Iron Condor, Straddle, Strangle, and hedging strategies. Developed Python backtesting workflows using Pandas and NumPy on 20+ years of historical market datasets to evaluate transaction-level performance, execution quality, strategy optimization, and risk management.

EDUCATION

Uttaranchal University

B.Tech - Computer Science and Engineering - CGPA: 8.5

2021 – 2025

Dehradun, Uttarakhand

Nirmal Ashram Deepmala Public School

Senior Secondary School - Percentage: 83

2020 – 2021

Rishikesh, Uttarakhand

Nirmal Ashram Deepmala Public School

High School - Percentage: 91.8

2018 – 2019

Rishikesh, Uttarakhand